

PHY301- Circuit Theory Mid Term Solved With References | By Masoom Fairy |

Phy301- Circuit Theory Solved Mid Term MCQS and Subjective with References.

Phy301-Mid Term Solved With References



By Masoom Fairy



Question No: 1 (Marks: 1) - Please choose one

If we connect 3 capacitors in series, the combined effect of all these capacitors will be

- ▶ equal to the sum of individual capacitors
- ▶ the sum of reciprocals of individual capacitors [Page 22](#)
- ▶ zero
- ▶ product of all

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Question No: 2 (Marks: 1) - Please choose one

A node which emerges as a result of combination of two ordinary nodes is called

- ▶ Heavy node
- ▶ Simple node
- ▶ Super mes
- ▶ **Super node** [Page 52](#)

Question No: 3 (Marks: 1) - Please choose one

Unit of inductance is

- ▶ mho
- ▶ ohm
- ▶ **Henry** [page 19](#)
- ▶ Farad

Question No: 4 (Marks: 1) - Please choose one

Reciprocal of resistance is

- ▶ Conductance
- ▶ Inductance
- ▶ **Capacitance** [Page 22](#)
- ▶ Resistivity

Question No: 5 (Marks: 1) - Please choose one

An ideal voltage source is a circuit element where the voltage across it is independent of

- ▶ resistance
- ▶ **current** [Page 26](#)
- ▶ capacitance
- ▶ inductance

Question No: 6 (Marks: 1) - Please choose one

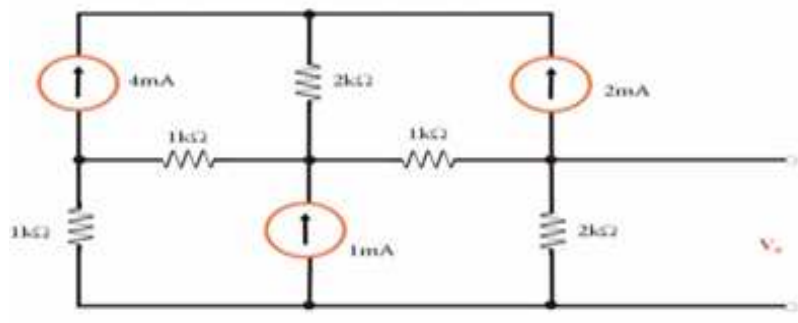
Sum of all currents entering in the node is equal to sum of currents leaving the node, is statement of

- ▶ Ohm's law
- ▶ **Kirchhoff's current law** [LECTURE 7](#)
- ▶ Jaime's law
- ▶ Rutherford's law

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Question No: 7 (Please choose one)

Which current source will be used for super mesh?



► 4mA

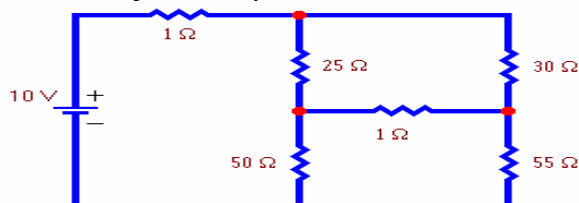
► 2mA

► 1mA when two meshes share current source then it became super mesh

► 2mA and 4mA

Question No: 8 - Please choose one

How many KVL equations can be written for the given circuit?



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► 2

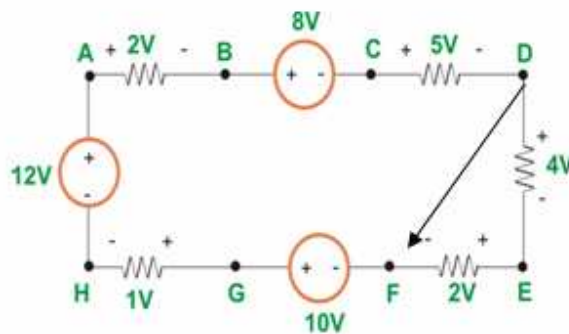
► 4

► 3 because 3 meshes or loops are formed and number of loops is equal to number of equations.

► 1

Question No: 9 - Please choose one

For the given circuit, voltage V_{DF} is



► 6v

► 2v

► 4v

► 10v

Question No: - Please choose one

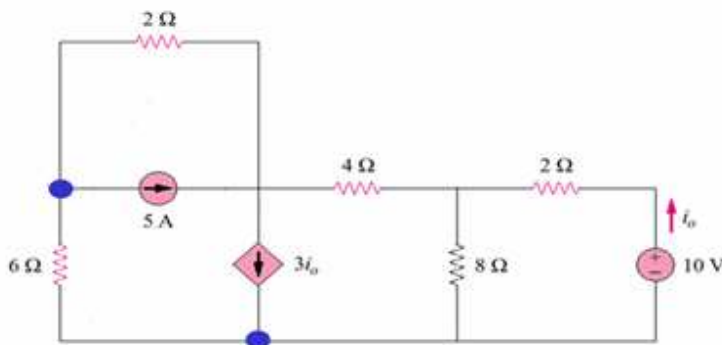
If a parallel circuit is open in main line, the current

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- ▶ increases in each branch
- ▶ is zero in all the branches
- ▶ is zero only in the branch that has highest resistance
- ▶ increases in the branch that has lowest resistance.

Question No: 11 - Please choose one

How many loop equations can be written for this circuit?



▶ 2

▶ 4

LOOP

It is the closed path for the flow of current in which no node is encountered more than once.

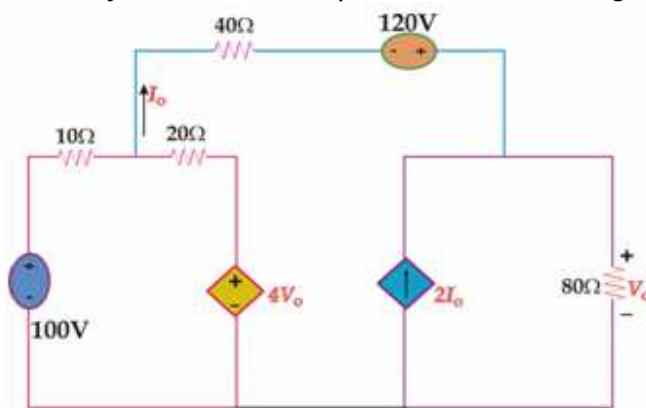
▶ 3

▶ 5

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Question No: 12) – Please choose one

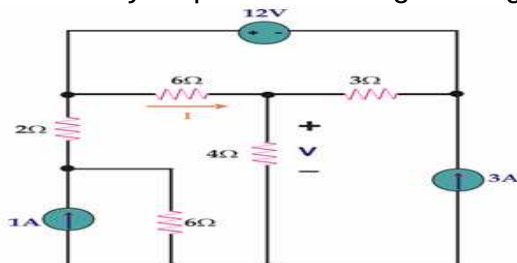
How many sources are dependent sources in given circuit?



- ▶ 3
- ▶ 2
- ▶ 4
- ▶ 1

Question No: 13 - Please choose one

How many loops can be assigned to given figure?



- ▶ 2

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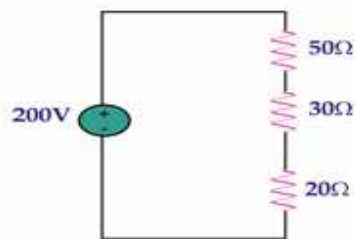
► 4

► 3

► 5

Question No: 14 - Please choose one

Through which resistance most current will flow



► 50Ω

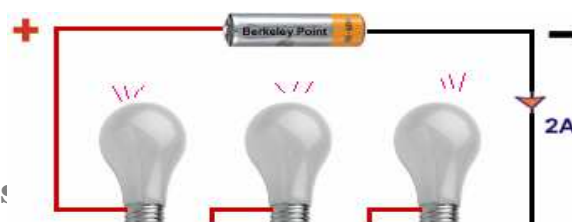
► 30Ω

► 20Ω

► same through all resistance Because same current flowing through series combination. Page 32

Question No: 15 - Please choose one

In given circuit power dissipated across each bulb is 20w. What would be the resistance of each bulb.



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► **5Ω**

$$P=V*I$$

$$20 = V*2A$$

$$20/2 = V$$

$$V=10v$$

So,

$$R=V/I = 10/2 = 5 \text{ ohm} = \text{Ans.....}$$

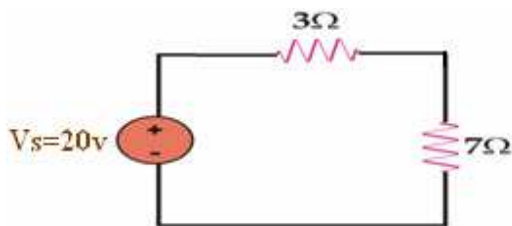
► 10Ω

► 40Ω

► 20Ω

Question No: 16 - Please choose one

In the figure below Voltage drop across 7Ω will be



► 20v

► **14v**

$$V=R/RT*VS$$

So,

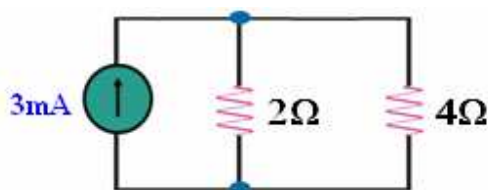
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$$V = 7/10 \times 20 = 14V$$

- ▶ 6v
- ▶ 10v

Question No: 17 - Please choose one

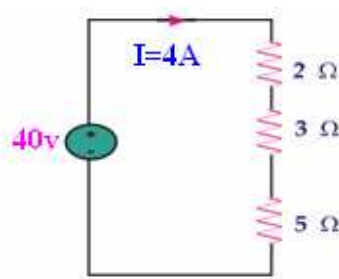
If current flowing through 4Ω is 1mA , voltage drop across it will be



- ▶ 2V
- ▶ 6V
- ▶ 4V not sure
- ▶ 12V

Question No: 18 - Please choose one

For the given figure current flowing through 5Ω is



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► 5A

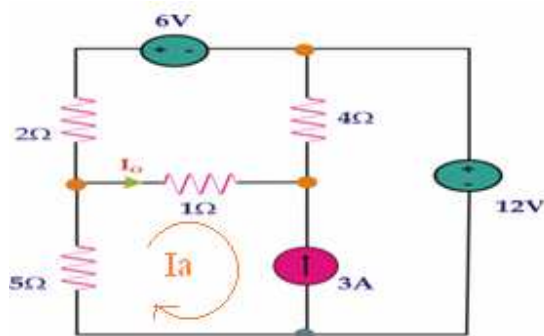
► 4A same current will flow in series circuit.

► 10A

► 2A

Question No: 19 - Please choose one

The value of I_a for given circuit is



► $I_a = I_o$

► $I_a = 3A$

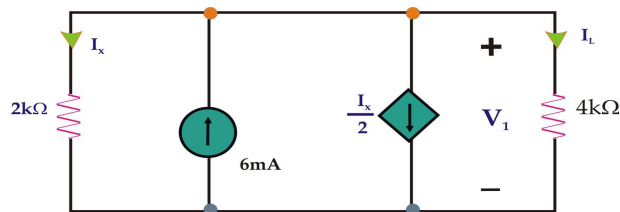
► $I_a = -3A$ Current is not in same direction with 3A.

► $I_a = 6V$

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Question No: 20 - Please choose one

In the given circuit the value of independent current source is



- I_x
- $I_x/2$ not sure
- 6mA
- I_L

MID TERM EXAMINATION SEMESTER Fall (2004) PHY- 301 CIRCUIT THEORY

(a) When placed close together, a positively charged material and a negatively charged material will

- a) Repel
- b) Become neutral
- c) Attract Page 9
- d) Exchange charges

II) Potential difference is another term for

- a) Energy

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b) Voltage Page 8

- c) Watt
- d) Distance of an electron from the nucleus.

III) There is no current in the circuit when

- a) A switch is closed
- b) A switch is open
- c) There is no voltage
- d) Answer (a) and (c)

e) Answer (b) and (c)

IV) How much resistance is required to limit the current from a 10V battery to 1Ma

- a) 100Ω
- b) $1.0\text{ k}\Omega$
- c) 10Ω

d) $10\text{k}\Omega$ $R=V/I$

So, $R=10/1=10\text{ ohm}$

V) Which of the following statements are true concerning the figure below

- a) R_1 and R_2 are in series with R_3 , R_4 , and R_5
- b) R_1 and R_2 are in series
- c) R_3 , R_4 , and R_5 are in parallel
- d) The series combination of R_1 and R_2 is in parallel with the series combination of R_3 , R_4 , and R_5 .

e) Answers (b) and (d)

**MID TERM EXAMINATION SEMESTER (Summer 2007) I
PHY- 301 CIRCUIT THEORY**

(b) Current divider is used when

- a) two or more resistances are in series to a voltage source
- b) two or more resistances are in parallel to a current source Sure**
- c) One resistance in parallel to current source
- d) None of above

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II) 10Ω and 5Ω resistances are in series. If current through 5Ω is 5 ampere , current of 10Ω will be

- a) 10A
- b) 20A
- c) 5A Page 32 Current will be same.**
- d) 40

III) If the current arrow is directed into the + marked terminal of an element, then $p=vi$ yields the.....

- a) absorbed power Not sure**
- b) emitted power
- c) loss power
- d) high power

IV) When a proton and an electron come close together they

- a) repel each other
- b) become neutral
- c) have no effect
- d) attract each other Page 9**

V) The resistance of an open circuit is

- a) high**
- b) low
- c) zero
- d) infinite

VI) The current passing through the conductor is directly proportional to the voltage provided temperature remains constant, is the statement of

- a) Kirchhoff's law
- b) Newton's law
- c) Ohm's law Page 24**
- d) Pascal law

MIDTERM EXAMINATION

Fall 2009

PHY301- Circuit Theory

Question No: 1 (Marks: 1) – Please choose one

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Which of the following statements is correct.

▶ **Neutrons and protons lie inside the nucleus.** Lecture 1 Page 4, 5, 6..

- ▶ Protons revolve around the nucleus.
- ▶ Neutron is +ve particle.
- ▶ Electron lies inside nucleus

Question No: 2 (Marks: 1) – Please choose one

Current through a short circuit is

- ▶ zero
- ▶ minimum
- ▶ **Maximum** Page 14 'Topic Short Circuit'.
- ▶ leakage current

Question No: 3 (Marks: 1) – Please choose one

When two resistances are connected in series

- ▶ They must both have same resistance value
- ▶ The voltage across each must be the same.
- ▶ They must have different resistance value.

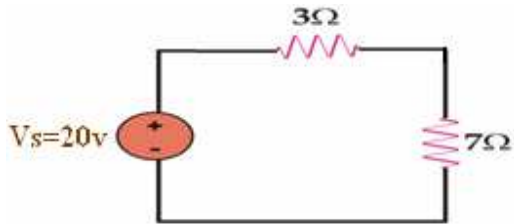
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► There is only one path for current for both resistances Sure

Question No: 4 (Marks: 1) – Please choose one

Using voltage divider in the figure below Voltage drop across 3Ω will be

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► 20v

► 14v

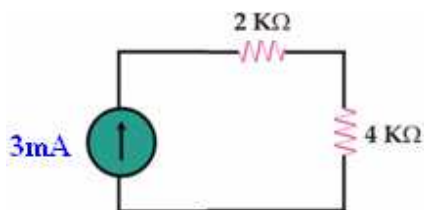
► 6v

$$V = \frac{R}{R_t} V_s \\ = \frac{3}{10} \times 20 = 6\text{v} = \text{Answer}$$

► 7v

Question No: 5 (Marks: 1) – Please choose one

In the given series fig. the total voltage of circuit is



► 6V

► 12V

► 18V

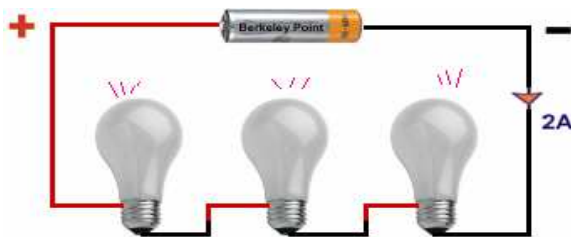
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$$\begin{aligned} V &= IR \\ &= 3\text{Ma} * 4+2 = 3\text{Ma} * 6 \\ &= 18\text{V} = \text{Ans....} \end{aligned}$$

► 10V

Question No: 6 (Marks: 1) – Please choose one

In given circuit power dissipated across each bulb is 20w. What would be the resistance of each bulb.



$$\begin{aligned} \text{► } 5\Omega \quad P &= V * I \\ 20 &= V * 2\text{A} \\ 20/2 &= V \\ V &= 10\text{v} \\ \text{So,} \\ R &= V/I = 10/2 = 5 \text{ ohm} = \text{Ans.....} \end{aligned}$$

► 10Ω

► 40Ω

► 20Ω

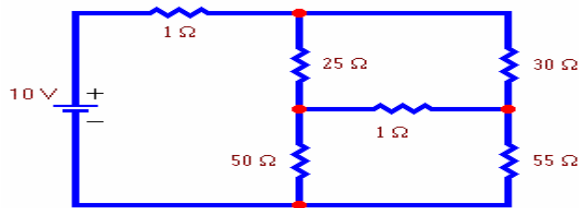
MID TERM FALL Winter 2012.

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Total Questions: 26:
20 MCQS & 6 Subjective Questions:

Question No: 1 - Please choose one

How many KVL equations can be written for the given circuit?



► 2

► 4

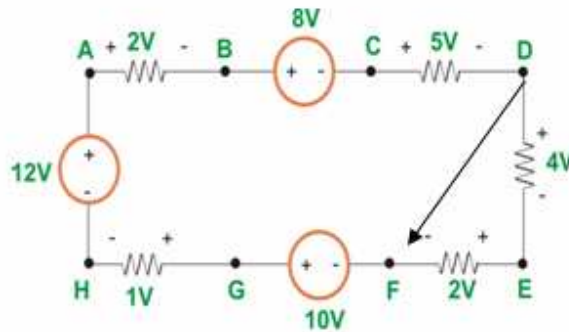
► 3 because 3 meshes or loops are formed and number of loops is equal to number of equations.

► 1

Question No: 2 - Please choose one

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For the given circuit, voltage V_{DF} is

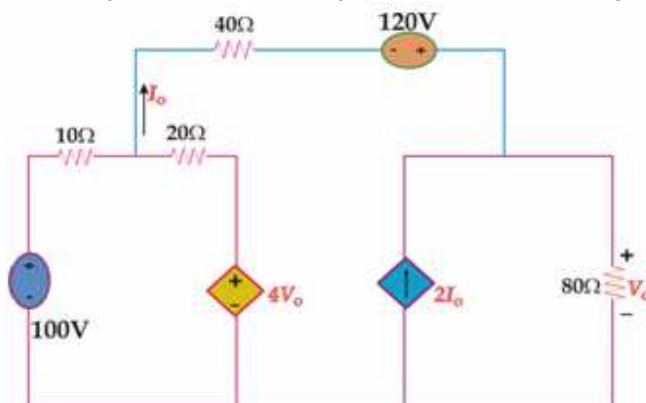


► 6v -2-4=0
So, $V_{df} = 6$ Answer

- 2v
- 4v
- 10v

Question No: 3 – Please choose one

How many sources are dependent sources in given circuit?



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► 3

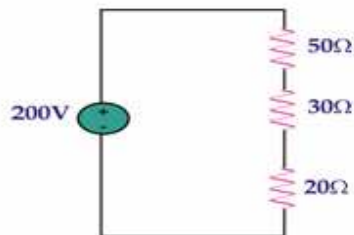
► 2

► 4

► 1

Question No: 4 - Please choose one

Through which resistance most current will flow



► 50Ω

► 30Ω

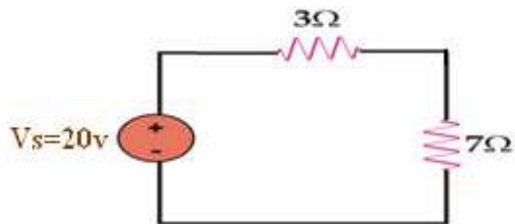
► 20Ω

► same through all resistance Because same current flowing through series combination. Page 32

Question No: 5 - Please choose one

In the figure below Voltage drop across 7Ω will be

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► 20v

► 14v

$$V = R/RT * VS$$

So,

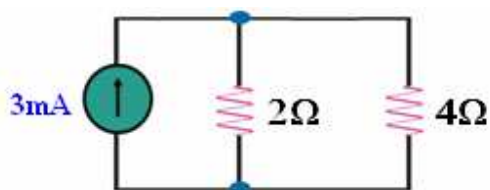
$$V = 7/10 * 20 = 14V$$

► 6v

► 10v

Question No: 6 - Please choose one

If current flowing through 4Ω is 1mA, voltage drop across it will be



► 2V

► 6V

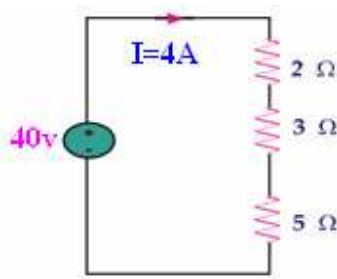
► 4V

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► 12V $V=IR \Rightarrow \frac{V}{R}=I$ So, $V=R/R_t \cdot V_t \Rightarrow 4/1.33 \cdot 4=12V=$ Answer.

Question No: 7 - Please choose one

For the given figure current flowing through 5Ω is



► 5A

► 4A same current will flow in series circuit.

► 10A

► 2A

Question No: 8 - Please choose one

If an atom has 8 electrons in its outer-most shell then the Valence of this atom is:

► 1

► Zero

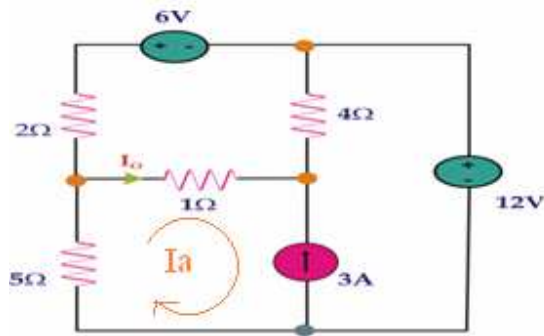
► 8

► 5

Question No: 9 - Please choose one

The value of I_a for given circuit is

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► $I_a = I_o$

► $I_a = 3A$

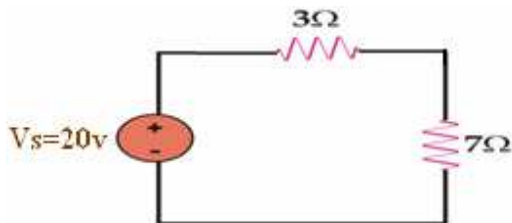
► $I_a = -3A$ Current is not in same direction with $3A$.

► $I_a = 6V$

Question No: 10 (Marks: 1) – Please choose one

Using voltage divider in the figure below Voltage drop across 7Ω will be

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► 20v

► **14v** $7/10 \times 20 = 14 = \text{Answer.}$

► 6v

► 7v

Question No: 7 (Marks: 2)

What is the unit of capacitance and Inductance?

Answer:

The SI unit of capacitance is the farad; 1 farad = 1 coulomb per volt.
The unit of inductance is the henry (H) named after American scientist and magnetic researcher Joseph Henry. 1 H = 1 Wb/A

Question No: 8 (Marks: 2)

Define KIRCHHOFF'S VOLTAGE LAW (KVL):

Answer:

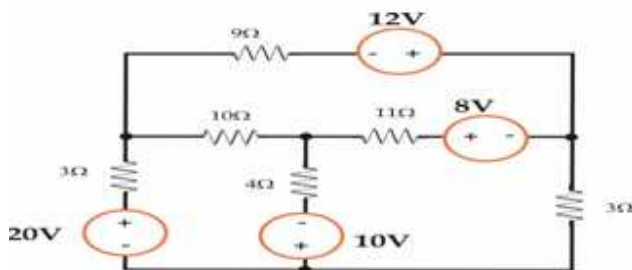
Kirchhoff's Voltage Law (or Kirchhoff's Loop Rule) is a result of the electrostatic field being conservative. It states that "the total voltage

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around a closed loop must be zero". If this were not the case, then when we travel around a closed loop, the voltages would be indefinite. So,
 $\sum V=0$

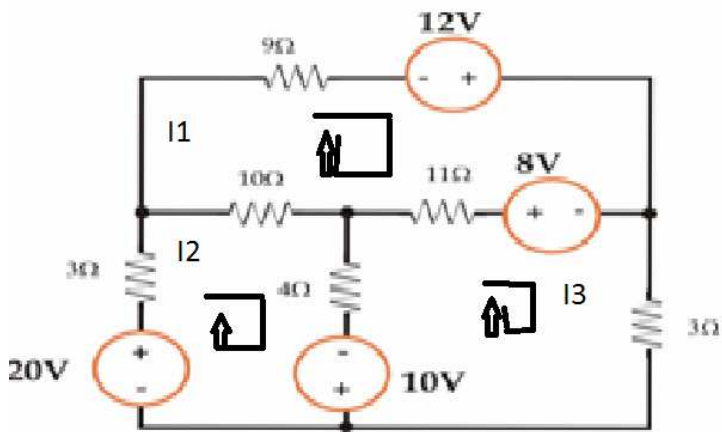
Question No: 9 (Marks: 3)

Draw and label loop currents for circuit.write kvl for any one loop equation.



Solution:

Circuit will be redrawn as



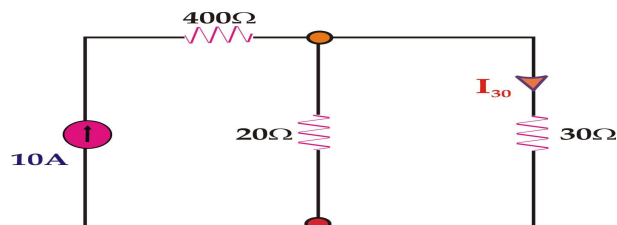
KVL for loop 1 is

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$$9I_1 + 1I_2 + 8 + 11(I_1 - I_3) + 10(I_1 - I_2) + 9I_1$$

Question No: 10 (Marks: 5)

You are given the circuit Use Current Division Rule directly to find I_{30} .



Solution:

Formula is

$$I_1 = R_2 / (R_1 + R_2)$$

AS we see that 30 ohm is in parallel with 20 ohm, So the resistance will become 12 ohm..

Now,

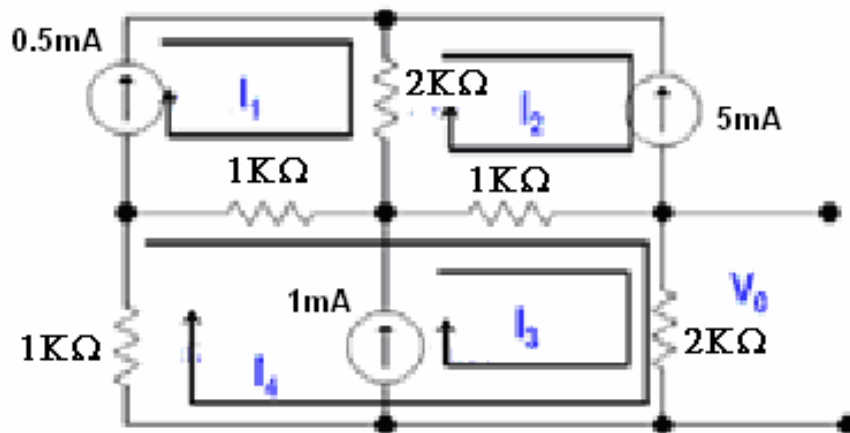
Using formula, we have

$$\begin{aligned} I_{30} &= \frac{400}{400 + 12} \times 10 \\ &= \frac{4000}{412} \\ &= 9.7A \end{aligned}$$

Question No: 11 (Marks: 5)

Determine the value I_4 for the given network.

PHY301- Circuit Theory Mid Term Solved With References [By Masoom Fairy]



Solution:

$$I_1 = 0.5mA,$$

$$I_2 = -5mA,$$

$$I_3 = 1mA$$

FOR I_4

$$1k(I_4 - I_1) + 1k(I_4 - I_2) + 1kI_3 + 2kI_3 + 1kI_4 = 0$$

$$= 2kI_4 - 1kI_2 + 3kI_3 = 0$$

PUTTING I VALUES

$$= 2kI_4 - 5 + 3 = 0$$

$$= 2kI_4 - 2 = 0$$

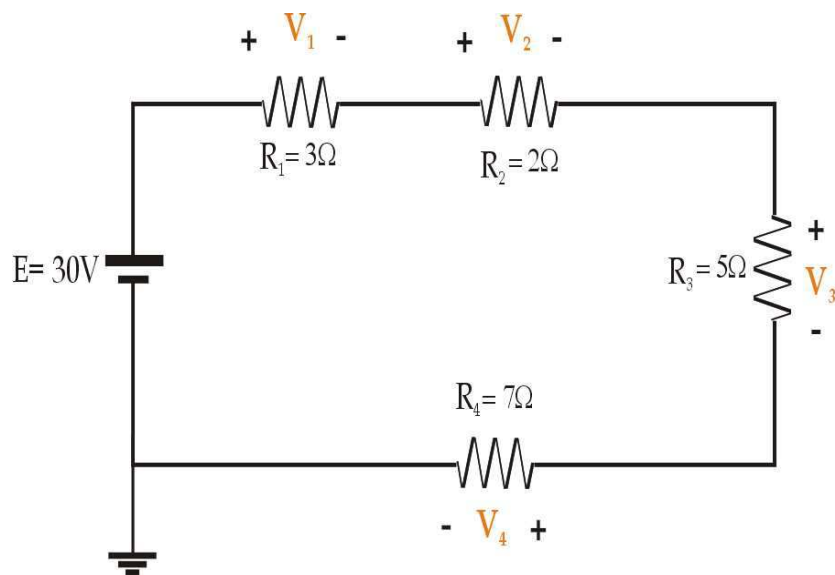
$$= I_4 = \frac{2}{2} = 1mA...$$

Question No: 12 (Marks: 10)

If 1.76A current is flowing through circuit

Calculate the power dissipated by R_1 , R_2 , R_3 and R_4 . Determine the power delivered by the source, Write each step of the calculation to get maximum marks and also mention the units of each derived value.

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Solution:

$$I=1.76 \text{ A}$$

As the formula for power delivered by source is

$$P = \frac{V^2}{R}$$

So,

$$P = \frac{30^2}{17} = \frac{900}{17}$$

$$P = 52.94 \text{ Watts}$$

Now Power dissipated in each resistor will:

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$$P = I^2 R$$

$$P_1 = (1.76)^2 \times 3$$

$$P_1 = 9.2 \text{ watts}$$

$$P_2 = (1.76)^2 \times 2$$
$$= 6.19 \text{ watts}$$

$$P_3 = (1.76)^2 \times 5$$
$$= 15.48 \text{ watts}$$

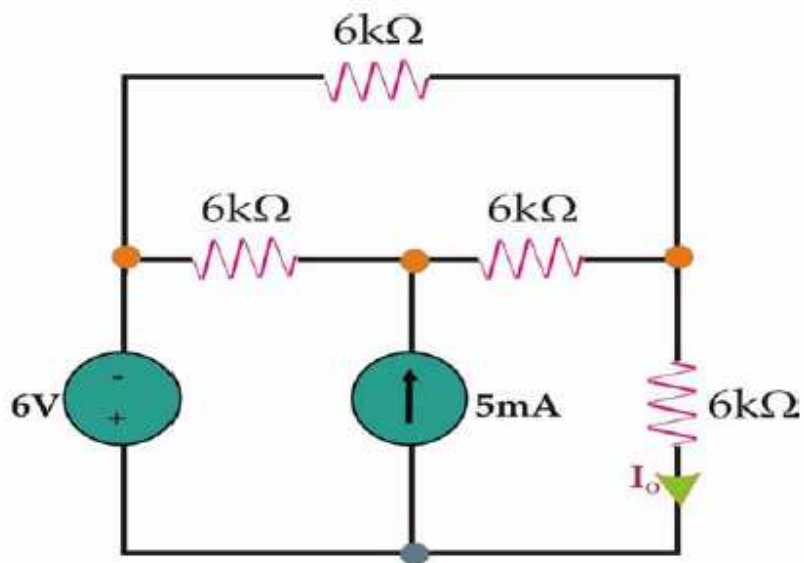
$$P_4 = (1.76)^2 \times 7$$
$$= 21.68 \text{ watts}$$

**MID TERM EXAMINATION
SEMESTER Fall (2004)I
PHY- 301 CIRCUIT THEORY**

Question:

Use nodal analysis to find I_o in the given network.. Identify and label each node otherwise you will lose your marks. Label circuit diagram properly.

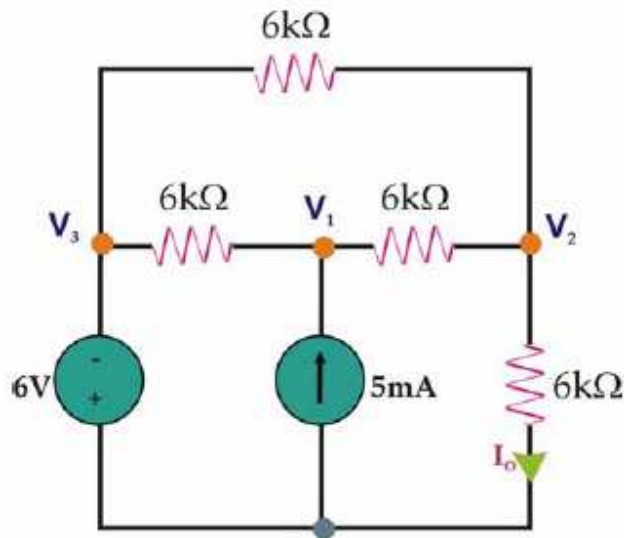
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Answer/Solution

We will redraw figure as

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At Node 3

$$V_3 = -6V$$

KCL at Node 1

$$(V_1 + 6)/6 + (V_1 - V_2)/6 = -5\text{mA} \text{ ----- (A)}$$

KCL at Node 2

$$(V_2 + 6)/6 + (V_2 - V_1)/6 + V_2/6 = 0 \text{ ----- (B)}$$

Solving (A) and (B) we get

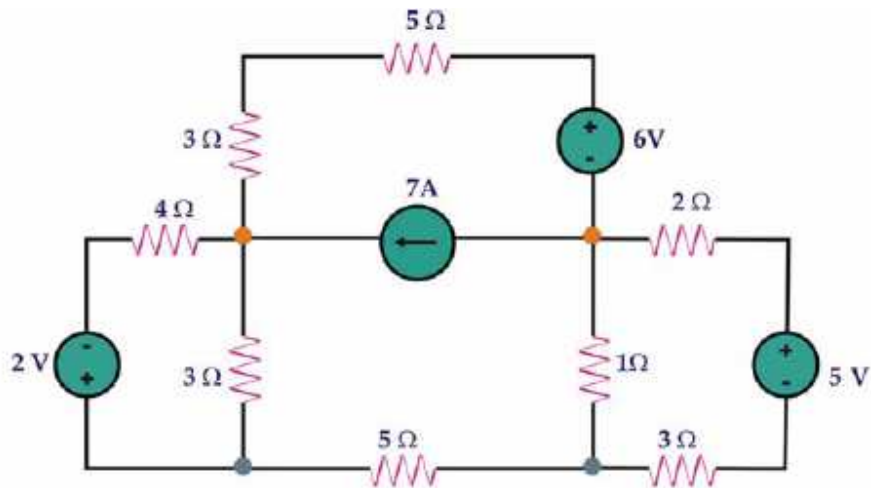
$$V_2 = 12/5 = 2.4$$

$$\text{Now } I_o = V_2/R = 2.4/6 = .4 \text{ Ma}$$

QUESTION:

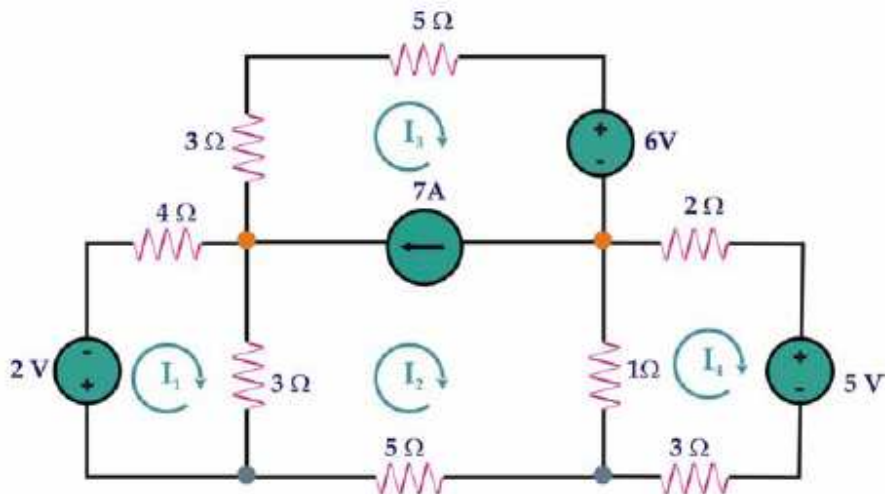
Use Mesh analysis to find currents through all Meshes. Identify and label each mesh otherwise you will lose your marks. Label circuit diagram properly.

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Solution:

_ We will redraw figure as:



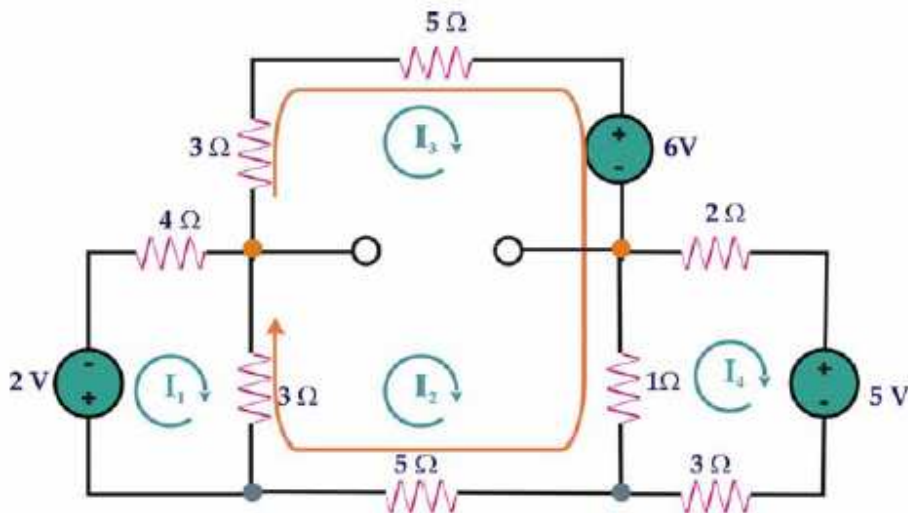
KVL equation for mesh 1

$$2 + 4I_1 + 3(I_1 - I_2) = 0$$

$$7I_1 - 3I_2 = -2 \text{ ----- (1)}$$

Now we will redraw figure for super mesh as

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KVL equation for super mesh

$$1(I_2 - I_4) + 8I_3 + 5I_2 + 6 + 3(I_2 - I_1) = 0$$

Simplifying above equation we get

$$-3I_1 + 9I_2 + 8I_3 - I_4 = -6 \dots \dots \dots (2)$$

KVL equation for Mesh 4

$$2I_4 + 5 + 3I_4 + 1(I_4 - I_2) = 0$$

Simplifying we get

$$-I_2 + 6I_4 = -5 \dots \dots \dots (3)$$

Constraint Equation

$$I_3 - I_2 = 7 \dots \dots \dots (4)$$

From Equation (1)

$$I_1 = (3I_2 - 2)/7$$

From Equation (3)

$$I_4 = (I_2 - 5)/6$$

From Equation (4)

$$I_3 = I_2 + 7$$

Putting the values of I_1 , I_3 , I_4 in equation (2) & simplifying we get

$$65I_2 = -2176$$

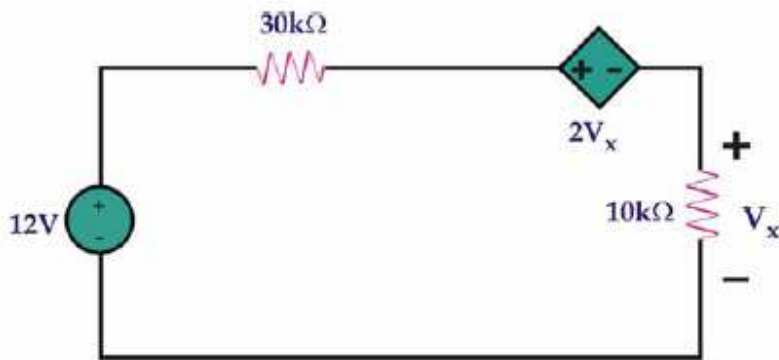
$$I_2 = -3.33A$$

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$$\begin{aligned} I_1 &= -1.71\text{A} \\ I_3 &= 3.67\text{A} \\ I_4 &= -1.38\text{A} \end{aligned}$$

QUESTION:

Find the power absorbed by the $30\text{k}\Omega$ resistor in the circuit below.



Solution:

Writing KVL equation

$$-12 - 30I + 2V_x + V_x = 0$$

$$-12 + 30KI + 20KI + 10KI = 0$$

$$I = 200\mu\text{A}$$

Therefore

$$P_{30k} = I^2 R$$

$$= (200 \times 10^{-6})^2 \times 30$$

$$= 1.2\text{MW}$$

QUESTION:

- (a) Two resistors R_1 and R_2 of 5Ω each and R_3 of 10Ω are in series. How much is R_T . If we applied 80V source across the series string, how much is the current in R_3 .
- (c) Define Potential difference. What is Conventional Current?

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Answer/Solution:

(d) Two resistors R_1 and R_2 of 5Ω each and R_3 of 10Ω are in series. How much is R_T . If we applied 80 V source across the series string, how much is the current in R_3 .

$$R_T = R_1 + R_2 + R_3 = 5 + 5 + 10 = 20\Omega \quad I = V_T / R_T = 80 / 20 = 4\text{ A}$$

Answer:

(e) Define Potential difference. What is Conventional Current?

Potential difference The potential difference is defined as the amount of work per charge needed to move electric charge from the second point to the first, or equivalently, the amount of work that unit charge flowing from the first point to the second can perform. In the SI system of units, potential difference, electrical potential and electromotive force are measured in volts, leading to the commonly used term voltage and the symbol V . OR Current flowing in a conductor is due to a potential difference between its ends. Electrons move from a point of less positive potential to more positive potential and the current flows in the opposite direction. The SI unit of potential difference is the volt (V). Volts are the measure of Potential Difference across circuit elements (battery, resistor etc.).

Conventional Current The motion of positive charge, in the opposite direction from electron flow, is considered as conventional current. Electricity was known of long before Benjamin Franklin. It was not understood very well, but it was known of. Scientists knew there were two kinds of electric charge. They knew there was electric current. Scientists believed that the opposite charges moved similarly in opposite directions. They defined one as positive and one as negative. They defined current to be in the direction of the positive charges. Later, they learned of their mistake. Only the negative charges move freely in conductors. Electrons had been defined as the negative charges. Current had been defined "backwards". It was too late to redefine all of electrical physics, so the inconvenience holds to this day. The direction that the electrons move is opposite the direction that current points. Because of how electricity works, it isn't much of a

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problem. Negative charge moving to the left through a wire has the same effect as positive charge moving to the right. So long as the total charge in the wire (protons and electrons) remains balanced, no trouble occurs.

Question: Marks: 2:

An electric heater takes 15A from the 120-V power line. Calculate the amount of power used also mention the units of derived value.

Solution:

$$P=VI$$

So,

$$P = 120 \times 15 = 1800 = \text{Answer}$$

MID TERM EXAMINATION SEMESTER (Summer 2007) I PHY- 301 CIRCUIT THEORY

Question:

Why does a person wearing rubber shoes do not get electric shock when touching a naked conducting wire.

Marks=3

Answer:

Rubber is a bad conductor of electricity (it used to be used to insulate wires), but a moderate conductor of heat. Experiment: cover the outside of a soup can with rubber bands, and place it inside a slightly larger can. Put a known amount of ice inside the soup can and see how long it takes to melt. Remove the rubber bands and try it again. Other substances (newspaper, aluminum foil, cardboard) can be used also for comparison.

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MID TERM FALL Winter 2012.

Subjective Paper:

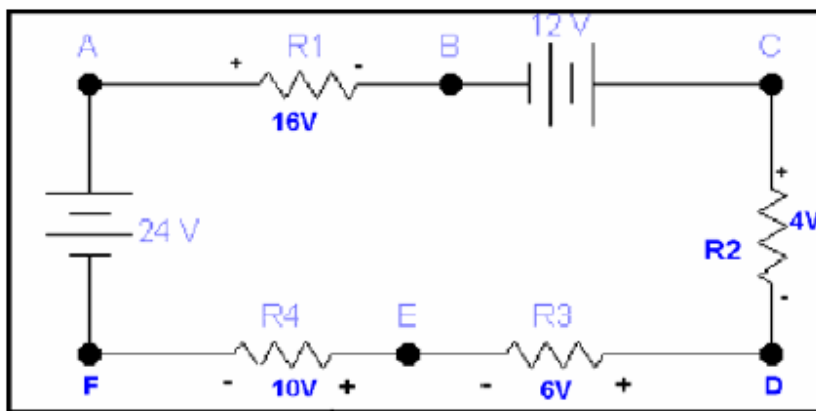
Question: 21: What will be the V(not) of 20 Ohm Resistance?

Marks: 2:

Answer: $I_2 = -4A$, $V=IR$ so, $-4A * 20 = -80 = \text{Answer}$.

Question: 22: Calculate the voltages V_{ae} .

Marks: 2:



Answer:

For path AEFA

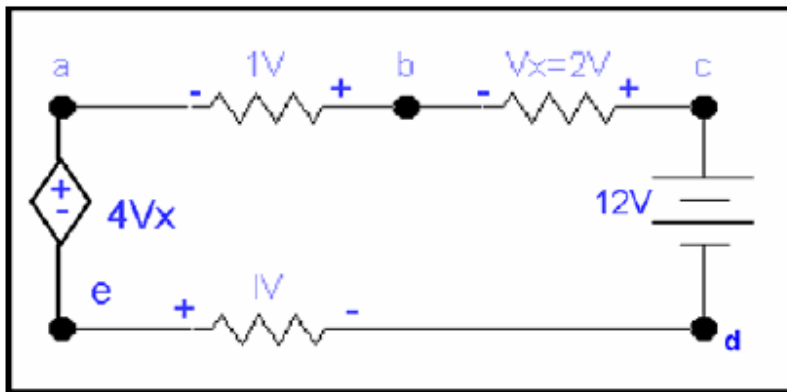
$$V_{AE} + 10 - 24 = 0 \text{ ----- (A)}$$

$$V_{AE} = 14 \text{ volt}$$

Question: 23: Calculate the voltage V_{ad} .

Marks: 3:

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Answer:

For Vad

There are two paths in which Vad is the only unknown quantity they are adea and adcba. For path adea the KVL equation will be

Vad

$$-1V - 4V_x = 0$$

where $V_x = 2V$

V

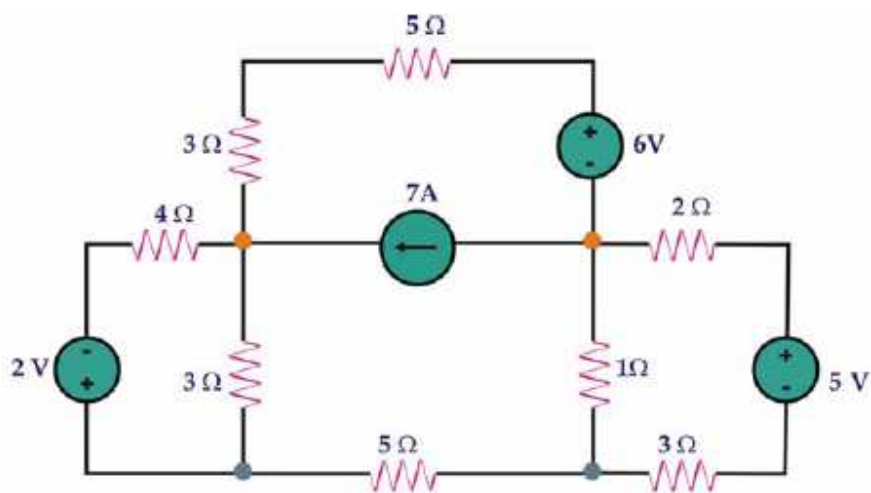
ad

$$-1 - 8 = 0$$

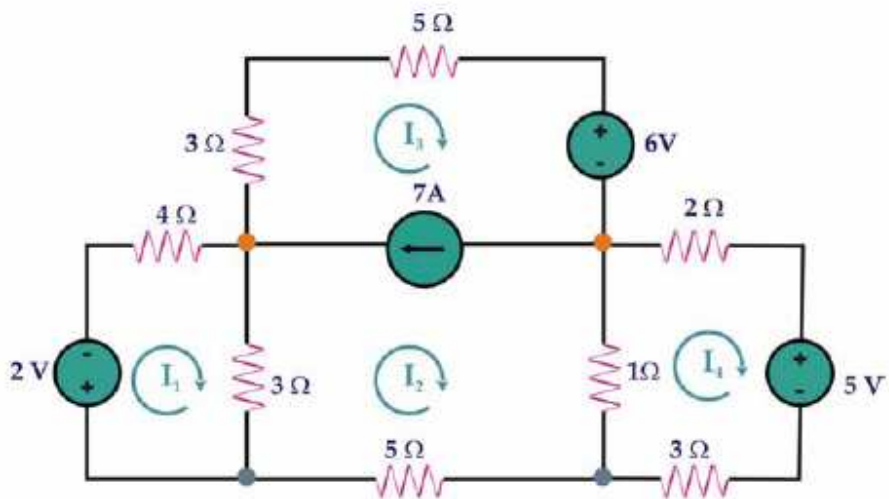
$$V_{ad} = 9 \text{ volts}$$

Question: 26: Use KVL labels each Mesh, write only Equation for Super Mesh. Marks: 5:

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Answer:



KVL equation for super mesh

$$1(I_2 - I_4) + 8I_3 + 5I_2 + 6 + 3(I_2 - I_1) = 0$$

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PHY301- CIRCUIT THEORY

Quiz#2 Solved with references

By Masoom fairy

Question # 1 of 10 (Start time: 12:12:50 PM) Total Marks: 1

A transformer may consists of

Select correct option:

one winding

two windings LECTURE 35

two or more than two windings

no winding

Question # 2 of 10 (Start time: 12:14:04 PM) Total Marks: 1

In a P- type semiconductor material, the majority carriers are

Select correct option:

Holes Lecture 29

Electrons

Ions

Protons

Question # 3 of 10 (Start time: 12:15:16 PM) Total Marks: 1

The primary and secondary winding of transformer are

Select correct option:

physically touched

physically isolated Lecture 35

touched with conductor

largely separated

Question # 4 of 10 (Start time: 12:16:12 PM) Total Marks: 1

Analysing the farward biase region of junction diode , the value of constant voltage, V_T is

Select correct option:

25.02V

10mV

25.02mV Lecture 31

5Mv

Question # 5 of 10 (Start time: 12:17:01 PM) Total Marks: 1

True or false, hole current is opposite of electron current

Select correct option:

true Lecture 29

false

Question # 6 of 10 (Start time: 12:18:10 PM) Total Marks: 1

Secondary voltage is greater in magnitude as compared to primary voltage in a

Select correct option:

step down transformer

step up transformer Lecture 35

isolated transformer

simple transformer

Question # 7 of 10 (Start time: 12:19:08 PM) Total Marks: 1

Which transformer gives less voltage at secondary as compared to primary

Select correct option:

isolated transformer

step up transformer

step down transformer Lecture 35

all transformer

Question # 8 of 10 (Start time: 12:19:46 PM) Total Marks: 1

A transformer will be out of phase when

Select correct option:

output voltage is 0 degree out of phase with input voltage

output voltage is 180 degree out of phase with input voltage Lecture 35

output voltage is 360 degree out of phase with input voltage

output voltage is same as input voltage

Question # 9 of 10 (Start time: 12:20:43 PM) Total Marks: 1

For a P-N junction under forward bias

Select correct option:

more forward current flows Not conform

no forward current flows
more reverse current flows
infinite reverse current flows

Question # 10 of 10 (Start time: 12:21:59 PM) Total Marks: 1

When the turn ratio of a transformer is 10 and primary ac voltage is 6v, the secondary voltage is

Select correct option:

36v

6v

60v

0.6v

$$N_2/N_1 = 10$$

$$V_1 = 6v$$

$$V_2 = ?$$

$$N_2/N_1 = V_2/V_1$$

$$N_2/N_1 * V_1 = V_2$$

$$10 * 6 = V_2$$

$$V_2 = 60v$$

CONTINUED:

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=1f27171e-e621-4b8c-a50b-4e68074048f4

BC110200184 : Faizan UI Hassan Syed

Time Left 81 sec(s)

Quiz Start Time: 11:20 AM

Question # 1 of 10 (Start time: 11:20:26 AM) Total Marks: 1

If turn ratio of a transformer is 5 and secondary current is 2A, the primary current is

Select correct option:

☐ 2.5A

☐ 10A

☐ 0.4A

☐ 7A

Click here to Save Answer & Move to Next Question

$$N_2/N_1 = 5$$

$$I_2 = 2A$$

$$I_1 = N_2/N_1 * I_2$$

$$= 5 * 2 = 10A = \text{Answer}$$

Lecture 35

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=80607a87-8bb9-4c6e-9720-cf9560a03249

BC110200184 : Faizan Ul Hassan Syed

Time Left 85 sec(s)

Quiz Start Time: 11:20 AM

Question # 2 of 10 (Start time: 11:21:08 AM) Total Marks: 1

Addition of impurities in semiconductor material to produce more current is called

Select correct option:

☐ intrinsic

☐ dopping

☐ bonding

☐ excitation

Click here to Save Answer & Move to Next Question

Dopping

Lecture 29

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=ecfaa302-c64f-4ad0-9108-e45359bb1dc3

BC110200184 : Faizan UI Hassan Syed

Time Left 86 sec(s)

Quiz Start Time: 11:20 AM

Question # 3 of 10 (Start time: 11:22:11 AM) Total Mark

There is a energy gap between valence and conduction band

Select correct option:

☐ forbidden

☐ permitted

Forbidden. Correct.

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Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=62a7852e-ef9d-43c7-a793-704a63b03659

BC110200184 : Faizan UI Hassan Syed

Time Left 85 sec(s)

Quiz Start Time: 11:20 AM

Question # 4 of 10 (Start time: 11:22:45 AM) Total Marks: 1

Diode small signal resistance r_d at bias current of 5mA for $n=1$, will be

Select correct option:

☐ 10 ohms

☐ 8 ohms

☐ 5 ohms

☐ 2.5 ohms

Click here to Save Answer & Move to Next Question

$$R_d = nV_t/I_d$$

$$= 1 \cdot 25/5 = 5 \text{ ohms}$$

LECTURE 34

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=0668d3ab-26da-4959-8faa-a843fdcc033b

BC110200184 : Faizan Ul Hassan Syed

Time Left 88 sec(s)

Quiz Start Time: 11:20 AM

Question # 5 of 10 (Start time: 11:23:10 AM) Total Marks: 1

There is no voltage drop across diode when forward biased, in case of

Select correct option:

☐ ideal diode model

☐ practical diode model

Click here to Save Answer & Move to Next Question

Ideal Diode model

LECTURE 33

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=f2283d85-6c53-4e9b-85cf-26ef92d82673

BC110200184 : Faizan Ul Hassan Syed

Time Left 85 sec(s)

Quiz Start Time: 11:20 AM

Question # 6 of 10 (Start time: 11:23:26 AM) Total Marks: 1

If 10w of power is applied to primary of an ideal transformer with a turn ratio of 5, the power delivered to the secondary load is

Select correct option:

☐ 25w

☐ 50w

☐ 0.5w

☐ 10w

Click here to Save Answer & Move to Next Question

50W I think.

According to formula:

$$P = V \cdot I$$

$$P_1 = V_1 \cdot I_1, P_2 = V_2 \cdot I_2$$

$$N_2/N_1 = V_2/V_1 = I_1/I_2$$

$$P_2 = N_2/N_1 \cdot P_1$$

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=04464e89-be25-4df9-9eff-7e707d16b4d3

BC110200184 : Faizan UI Hassan Syed

Time Left 86 sec(s)

Quiz Start Time: 11:20 AM

Question # 7 of 10 (Start time: 11:24:05 AM) Total Marks: 1

The width of the depletion layer of a junction

Select correct option:

- ☐ decreases with light doping
- ☐ increases with heavy doping
- ☐ under reverse bias
- ☐ is increased under reverse bias

Click here to Save Answer & Move to Next Question

Is increased under reverse bias.

LECTURE 43 (out of 29 to 36)

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=b94eff14-f78d-4619-a2e0-afc4e46dc2ea

BC110200184 : Faizan Ul Hassan Syed

Time Left 86 sec(s)

Quiz Start Time: 11:20 AM

Question # 8 of 10 (Start time: 11:24:24 AM) Total Marks: 1

Using exponential model,for a diode that conducts at 1mA with $n=2$, the change in current due to change of voltage by 5mv will be

Select correct option:

☐ 0.4mA

☐ 0.8mA

☐ 3mA

☐ 0.11mA

Click here to Save Answer & Move to Next Question

0.11mA (not conform).

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=e734801f-b5d7-4f57-9988-be8fd94b4185

BC110200184 : Faizan Ul Hassan Syed

Time Left 85 sec(s)

Quiz Start Time: 11:20 AM

Question # 9 of 10 (Start time: 11:25:03 AM) Total Marks: 1

In an N- type semiconductor material , the majority carrier are

Select correct option:

☐ holes

☐ electrons

☐ ions

☐ protons

Click here to Save Answer & Move to Next Question

Electrons.

LECTURE 29

Quiz - Mozilla Firefox

quiz.vu.edu.pk/QuizQuestion.aspx?ver=941083a7-edb2-426a-8ed4-1369f82b2ada

BC110200184 : Faizan Ul Hassan Syed

Time Left 86 sec(s)

Quiz Start Time: 11:20 AM

Question # 10 of 10 (Start time: 11:25:19 AM) Total Marks: 1

If a silicon diode conducts at 2mA and whose $n=1$, the slope of diode will be at

Select correct option:

☐ 1/25 ohms

☐ 1/2 ohms

☐ 2/25 ohms

☐ 2 ohms

Click here to Save Answer & Move to Next Question

2/25 ohms. CORRECT

$$\text{Slop} = 1/R_d$$

$$R_d = I / nV_t$$

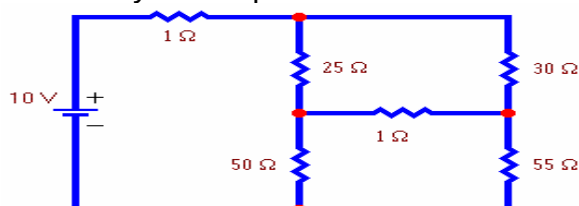
$$= 2/1 * 25 = 2/25 \text{ ohms.}$$

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**Total Questions: 26:
20 MCQS & 6 Subjective Questions:
10 MCQS which I remember are given below:**

Question No: 1 - Please choose one

How many KVL equations can be written for the given circuit?



- ▶ 2
- ▶ 4

Page | 1

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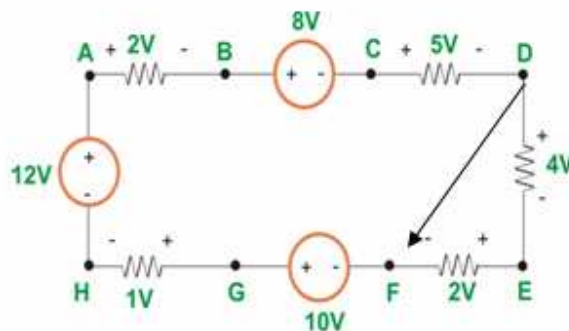
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► 3 because 3 meshes or loops are formed and number of loops is equal to number of equations.

► 1

Question No: 2 - Please choose one

For the given circuit, voltage V_{DF} is



► 6v $-2-4=0$
So, $V_{df}= 6$ Answer

► 2v

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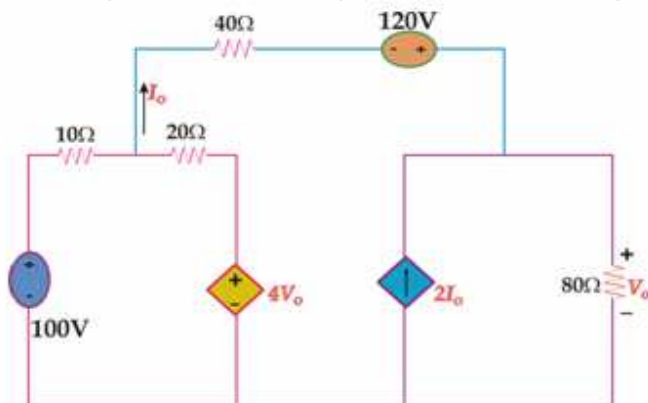
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- 4v
- 10v

Question No: 3 – Please choose one

How many sources are dependent sources in given circuit?



- 3
- 2

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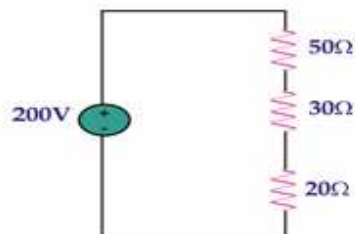
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► 4

► 1

Question No: 4 - Please choose one

Through which resistance most current will flow



► 50Ω

► 30Ω

► 20Ω

► same through all resistance Because same current flowing through series combination. Page 32

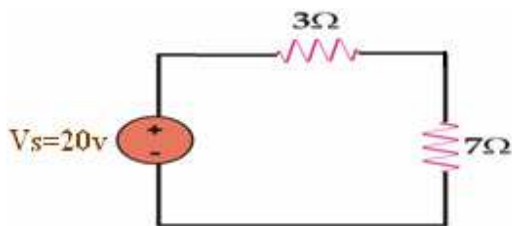
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Question No: 5 - Please choose one

In the figure below Voltage drop across 7Ω will be



► 20v

► 14v

$$V = R/RT \cdot V_S$$

So,

$$V = 7/10 \cdot 20 = 14V$$

► 6v

► 10v

Question No: 6 - Please choose one

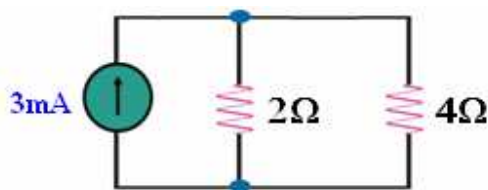
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If current flowing through 4Ω is 1mA , voltage drop across it will be



► 2V

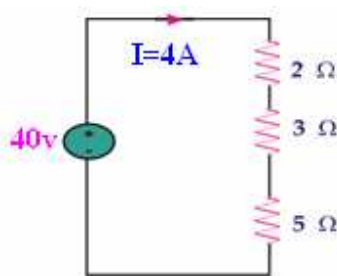
► 6V

► 4V

► 12V $V=IR \Rightarrow \frac{1}{4}=4$ So, $V=R/R_t \cdot V_t \Rightarrow 4/1.33 \cdot 4=12\text{V}$ = Answer.

Question No: 7 - Please choose one

For the given figure current flowing through 5Ω is



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► 5A

► **4A** same current will flow in series circuit.

► 10A

► 2A

Question No: 8 - Please choose one

If an atom has 8 electrons in its outer-most shell then the Valence of this atom is:

► 1

► **Zero**

► 8

► 5

Question No: 9 - Please choose one

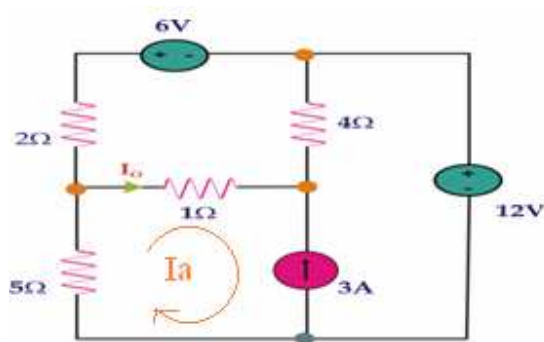
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The value of I_a for given circuit is



- ▶ $I_a = I_o$
- ▶ $I_a = 3A$
- ▶ $I_a = -3A$ Current is not in same direction with 3A.
- ▶ $I_a = 6v$

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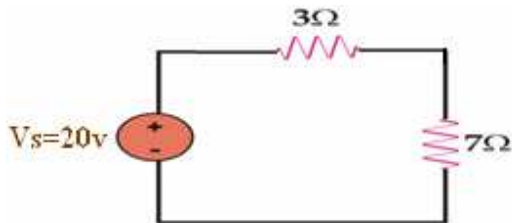
Question No: 10 (Marks: 1) – Please choose one

Using voltage divider in the figure below Voltage drop across 7Ω will be

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► 20v

► **14v** $7/10 \times 20 = 14 = \text{Answer.}$

► 6v

► 7v

Subjective Paper:

Question: 21: What will be the V(not) of 20 Ohm Resistance?

Marks: 2:

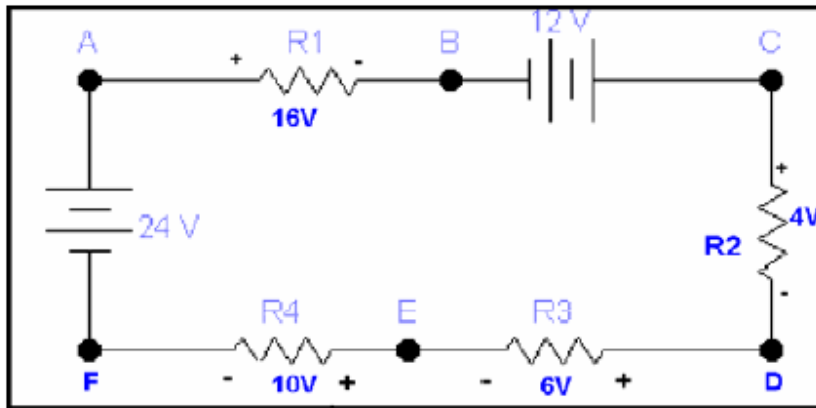
Answer: $I_2 = -4A$, $V=IR$ so, $-4A \times 20 = -80 = \text{Answer.}$

Question: 22: Calculate the voltages Vae.

Marks: 2:

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Answer:

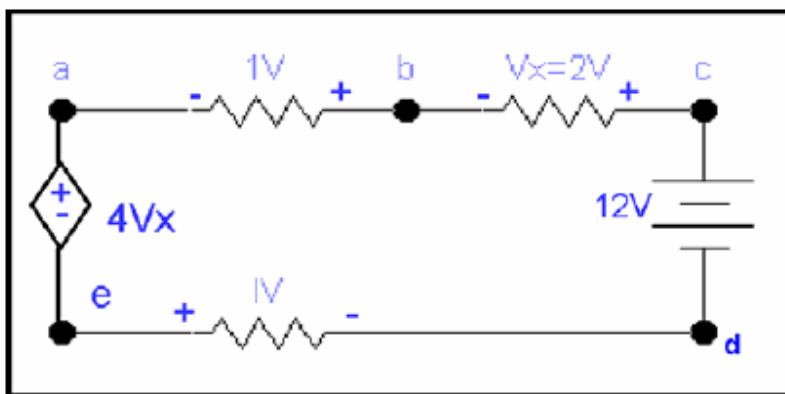
For path AEFA

$$V_{AE} + 10 - 24 = 0 \text{ ----- (A)}$$

$$V_{AE} = 14 \text{ volt}$$

Question: 23: Calculate the voltage V_{ad} .

Marks: 3:



Answer:

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For Vad

There are two paths in which Vad is the only unknown quantity they are adea and adcba. For path adea the KVL equation will be

Vad

$$\mathbf{-1V - 4V_x = 0}$$

where $V_x = 2V$

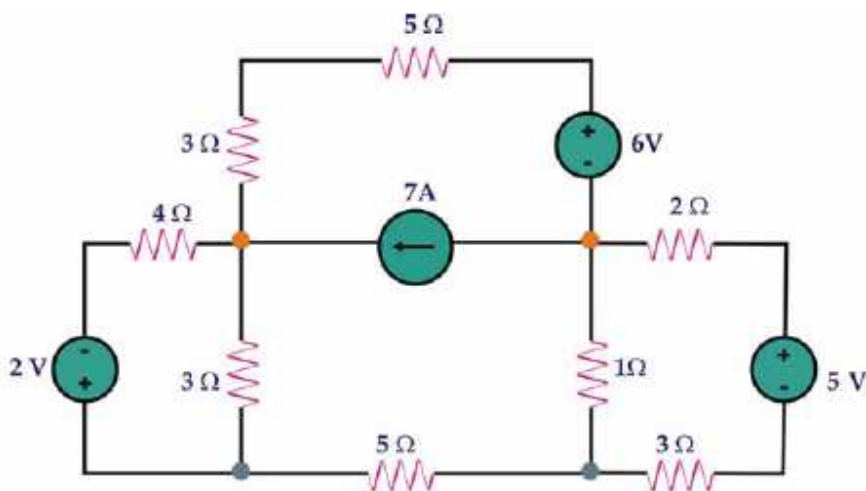
V

ad

$$\mathbf{-1 - 8 = 0}$$

$$\mathbf{V_{ad} = 9 \text{ volts}}$$

Question: 26: Use KVL labels each Mesh, write only Equation for Super Mesh. Marks: 5:



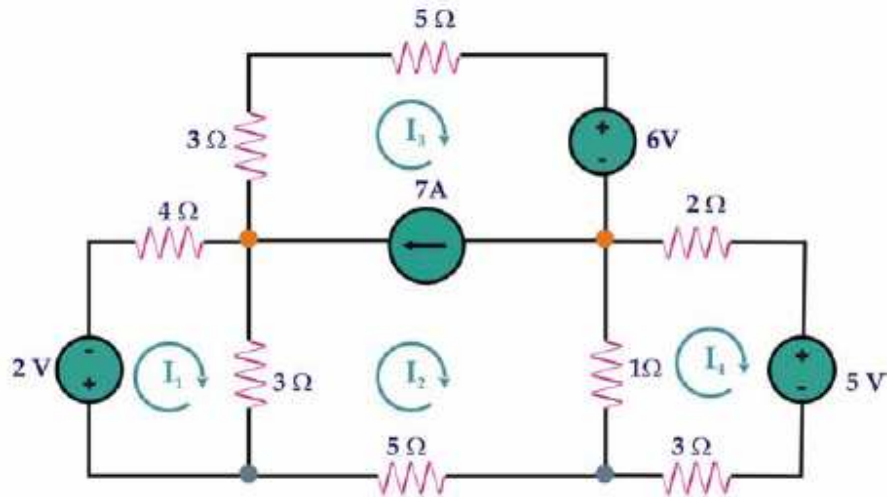
Answer:

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KVL equation for super mesh

$$1(I_2 - I_4) + 8I_3 + 5I_2 + 6 + 3(I_2 - I_1) = 0$$

Note:

Question 24 and 25 are not remember but they both are from KVL lectures.